

Headline statements of the grammar §4 formalisation

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Abstract

Astra #12 rank 5 (“unusually high value at 151 units”). `Laplace/Grammar/Headline.lean` collects thin, paper-facing wrappers of the main theorems with all hypotheses spelled out, a module `docstring` fixing the scope ($d = 2$, one chart, Taylor data in the ρ -weighted ℓ^1 space, canonical coefficients), and an explicit not-claimed list. It also identifies the Lean chart integral with the paper’s standard integral (`headline_standard_integral`). Nothing is re-proved. Zero `sorry/axiom`.

1 The headline statements

1. `headline_standard_integral`: $\text{twoDAmp } \beta b N h k (\text{anaAmp } (\text{ampCoeff } \beta x y) b) = \iint_{(0,b]^2} u^{h_1} v^{h_2} e^{-\beta(Nu^{k_1}v^{k_2})^2} \eta(u,v) e^{\beta Nu^{k_1}v^{k_2} \xi(u,v)}, \xi = \sum x_{ij} u^i v^j, \eta = \sum y_{ij} u^i v^j$ (defn:std_integral at $\sqrt{n} = N$).
2. `headline_taylor_tree`: `thm:TaylorTree` for $d = 2$, all (h, k) , every truncation T (unit 129).
3. `headline_taylor_tree_uniform`: $\text{remainder} \leq K(\beta, b, \rho, M, h, k, T) N^{-2T} (1 + \log N)$ uniformly on $\|a\| \leq M$ (units 135, 149).
4. `headline_leading_coefficient`: $A_p = \frac{y_{00}}{k_1 k_2} \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta s x_{00}} ds$ and $y_{00} > 0 \Rightarrow A_p > 0$ (units 124, 154).
5. `headline_taylor_data`: $\partial_v^j \partial_u^i \xi(0, 0) = i! j! x_{ij}$ (unit 144).
6. `headline_expectation`: averaged Taylor tree with uniform remainder, only joint measurability assumed (units 137–139).
7. `headline_coefficient_lipschitz`: local Lipschitz continuity of A_α in the Taylor data (unit 145).
8. `headline_coefficients_in_distribution`: finite coefficient vectors converge in distribution (unit 147).
9. `headline_normalised_remainders`: ordered A/B-slot normalised remainders converge in distribution for measurable Taylor data converging in distribution (units 148–153).
10. `headline_far_phase`: $|\int_E \eta e^{-\beta n K + \beta \sqrt{n K} \psi}| \leq (\int_E |\eta|) e^{-\beta n \varepsilon / 2 + \beta M^2 / 2}$ (unit 151).

2 Scope and what is not claimed

Scope: dimension two, one resolution chart, box $[0, b]^2$, monomial phase $u^{k_1}v^{k_2}$ with Jacobian exponents (h_1, h_2) , evaluation at $\sqrt{n} = N$, phase and amplitude given by Taylor data absolutely summable at radius $\rho > b$, canonical (cutoff-independent) coefficients. Not claimed: general d ; the multi-chart/resolution assembly; the functional CLT for the empirical process; the identification of the paper's C^ω topology with the weighted coefficient norm (Cauchy estimates); convergence of expectations from convergence in distribution beyond the stated domination hypotheses; uniform far-phase decay without control of the random bound M ; a full posterior-ratio expansion (only the positivity of the leading denominator is proved so far).

3 Lean notes

- The standard-integral identification is two nested `setIntegral_congr_funs` with `anaAmp_ampCoeff` on $\text{Ioc} \subseteq \text{Icc}$.
- Wrappers are proved by direct application; the file compiled on the first attempt apart from two over-long docstring lines.
- Section-scoped `variables` separate the two probabilistic settings (varying sample spaces for the continuous-mapping transfer; a fixed sample space with a countably generated filter for the Slutsky-based normalised remainders).